Home Network Playbook

Maximizing home infrastructure: document it as found, then improve it. October 2026.

The pitch

Most homes already contain structured wiring that goes unused. This house has **Cat6e and RG6 to every room**, all home-run to one garage comm panel, yet remote rooms are served over a single shared MoCA 2.0 channel through a 4-way splitter, behind a double-NAT router stack. This playbook records the setup so others can replicate it and lays out a measured path to use the wiring properly.

Phase 1: As found

Layer	What is there
WAN	AT&T Fiber to AT&T ONT on exterior wall; Cat6 to garage patch panel (B1), patched to the BGW210 run (B2)
Media room	AT&T BGW210 (default, Wi-Fi off) → Google Wifi hub AC-1304 → TP-Link switch → Actiontec ECB6200 MoCA
Coax plant	GE 4-way splitter, 5-2500 MHz, IN unused; 4 active legs: media room, sitting room cabinet, 2nd floor back right, master
Remote rooms	ECB6200 → Google Wifi node (WAN port); mesh backhaul over MoCA
Cat6e	12-port keystone patch panel; RJ45 at panel; only 2 rooms RJ45 at room end; 5 ports unidentified
Unused	All other RG6 runs (not connected at either end)

Signal path

Fiber → ONT → panel → **BGW210** → **Google Wifi hub** → **TP-Link switch** → **MoCA** → RG6 → panel splitter → **3 x (MoCA → Google Wifi node)**

Phase 2: Improvement opportunities

ID	Opportunity	Status	Impact
P2-0	Identify and label the 5 unknown patch-panel ports	To do	High
P2-1	Re-terminate remaining room-end Cat6e from POTS to RJ45	Panel + 2 rooms done	Very high
P2-2	Move gateway + hub to the garage panel, add a switch, patch rooms to it	Option C chosen	Very high
P2-3	Cap the splitter IN port with a 75-ohm terminator; check adapter	To do	Medium
P2-4	Remove double NAT: BGW210 IP Passthrough to the hub	To do	High
P2-5	Move mesh backhaul from MoCA to Cat6 where a run exists	To do	High
P2-6/7	Label and document everything; UPS for ONT, gateway, switch	Partly done	Medium

Measure, do not assume

Measured so far (upload): gateway 960 Mbps, wired over MoCA 890 Mbps, Wi-Fi 256 Mbps, so Wi-Fi is the weakest link. Baseline every room with iperf3 before changes (expect about 940 Mbps over gigabit Ethernet), confirm double NAT with a traceroute, read MoCA link rates per node, and confirm mesh points report wired backhaul. Rollback is simple at each step because nothing is cut; old plugs and MoCA adapters stay until the Cat6 path is proven.

Replicate it

Full schematics (Mermaid), patch panel map, step-by-step procedures, inventory template and glossary are in the repository docs: **01-as-found**, **02-improvements**, **03-inventory-template**, **04-verification**, **05-glossary**.